BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Sequence, Input and Output

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Name one operating-system function.

2. What does software copyright restrict?
   A) Using any keyboard
   B) Learning to code
   C) Copying without permission
   D) Saving your own work

3. What is a typical feature of proprietary software?

4. A school needs a large, cheap archive drive. A pupil's only coursework copy is on an SSD. Correct both claims. Explain why. (a) An HDD is always faster and tougher than an SSD. (b) An SSD never fails, so needs no backup.

Today's objectives

Write programs containing instructions executed in sequence. Use sequence and input in a practical scenario.

Key vocabulary - Read, cover, recall

Sequence
Instructions execute in order, so each value is ready before it is used.

Input
The input() command obtains data from a user and can store it in a variable.

Processing
Expressions calculate a new value from existing values.

Output
print() displays the result of a completed calculation.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

What is Sequence?

Sequence - Instructions execute in order, so each value is ready before it is used.

In this example

A bus ticket price app uses input; The input() command obtains data from a user and can store it in a variable.

Follow the effect

Expressions calculate a new value from existing values. Explain how this changes the eventual program result.

Check understanding

Output is used for a specific reason: print() displays the result of a completed calculation.

I Do - Worked model

A bus ticket price app is being written. Explain how sequence and input are used to make the program work. [4 marks]

- We are designing a bus ticket price app. Identify which data and decisions are essential.
- Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1].
- Input [1] - The input() command obtains data from a user and can store it in a variable. [1].
- Two linked points for each technique give four mark points.

We Do - Guided practice

1. In a school uniform total, explain how sequence helps the program. [2 marks]

2. For a two sports scores, explain why input matters. [2 marks]

3. When designing a cinema booking, explain how processing and output should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write one line to display the name the user enters. Assume the input is "Ava".

```
name = input("Your name?")
```

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A shop receipt needs to work correctly. Explain how input and processing would help. [4 marks]

Q4. A pupil is programming a online café bill. Explain the roles of sequence, processing and output in their solution. [6 marks]

Exit Ticket - True or False?

Instructions execute in order, so each value is ready before it is used.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Memory management, user management, file management, a user interface or peripheral management.

2. C - Copying without permission

3. Source code is normally controlled by its owner and users receive specified usage rights.

4. (a) An HDD can give lots of space at low cost per GB. Check the device and task. (b) An SSD can fail through faults or wear, so keep backups. No device is best at everything.

I Do - Explain the model

We are designing a bus ticket price app. Identify which data and decisions are essential.

Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1].

Input [1] - The input() command obtains data from a user and can store it in a variable. [1].

Two linked points for each technique give four mark points.

We Do - Model answers

1. Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1].

2. Input [1] - The input() command obtains data from a user and can store it in a variable. [1].

3. Processing [1] - Expressions calculate a new value from existing values. [1]. Output [1] - print() displays the result of a completed calculation. [1].

Code First - Answer

```
name = input("Your name?")
print(name)
```

Expected output: Ava (assuming input Ava)

You Do - Self-mark

Q2: Sequence [1]; Processing [1].

Q3: Input [1] - The input() command obtains data from a user and can store it in a variable. [1]. Processing [1] - Expressions calculate a new value from existing values. [1].

Q4: Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1]. Processing [1] - Expressions calculate a new value from existing values. [1]. Output [1] - print() displays the result of a completed calculation. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.